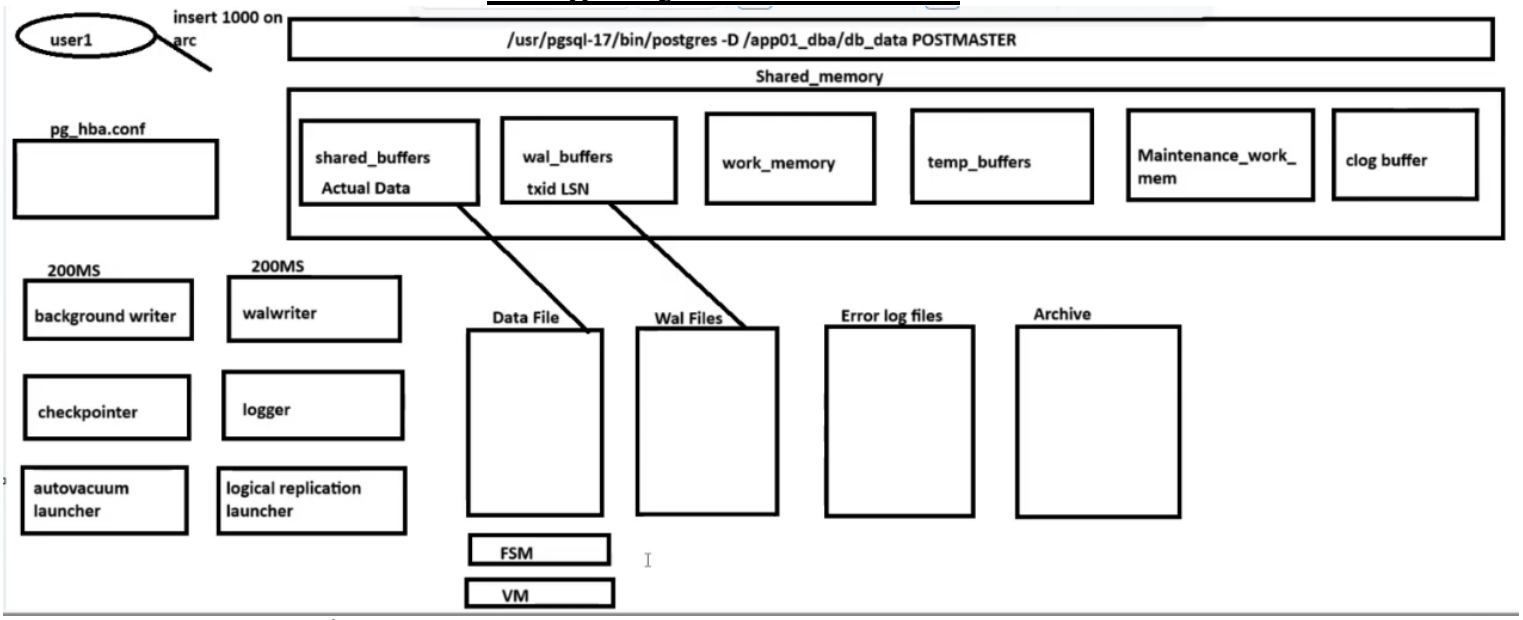
PostgreSQL Architecture




```
dba@example~]$ps -ef|greppostgres
dba         3956    2245  007:09 ?        00:00:00 /usr/pgsql-17/bin/postgres -D
/app01_dba/db_data
dba         3957    3956  007:09 ?        00:00:00 postgres: logger
dba         3958    3956  007:09 ?        00:00:00 postgres: checkpointer
dba         3959    3956  007:09 ?        00:00:00 postgres: background writer
dba         3961    3956  007:09 ?        00:00:00 postgres: walwriter
dba         3962    3956  007:09 ?        00:00:00 postgres: autovacuum launcher
dba         3963    3956  007:09 ?        00:00:00 postgres: logical replication launcher
dba         4055    3775  0 07:09 pts/0    00:00:00 grep --color=auto postgres
[dba@example~]$psql -p 5432 -U dba postgres
psql (17.4)
Type "help" for help.
postgres=# create table arc(id integer,namevarchar(20));
CREATE TABLE
postgres=# \d+ arc
                                                  Table "public.arc"
 Column |         Type          | Collation | Nullable | Default | Storage  | Compression | Stats target |
Description
--------+-----------------------+-----------+----------+---------+----------+-------------+--------------+---
----------
 id     | integer               |           |          |         | plain    |             |              |
 name   | character varying(20) |           |          |         | extended |             |              |
Access method: heap
postgres=# select pg_relation_filepath('arc');
pg_relation_filepath
-----------------------
 base/5/32789
(1 row)
postgres=# \q
[dba@example~]$ cd
.cache/     db_data/    db_wal/     postgresql/
[dba@example~]$ cd db_data/base/5/
[dba@example5]$ ls -lrth 32789
-rw-------. 1 dba dba 0 Mar 14 07:23 32789
[dba@example5]$ ls -lrth 32789
-rw-------. 1 dba dba 440K Mar 14 07:31 32789
[dba@example5]$psql -p 5432 -U dba postgres
psql (17.4)
```

```
[dba@example~]$psql -p 5432 -U dba postgres
psql (17.4)
Type "help" for help.
postgres=# \dt
        List of relations
 Schema |  Name  | Type  | Owner
--------+--------+-------+-------
 public | arc    | table | dba
 public | orders | table | dba
 public | tab1   | table | dba
 public | tb1    | table | dba
 public | test1  | table | dba
(5 rows)
postgres=# begin;
BEGIN
postgres=*# insert into arc values(generate_series(1,10000),'manu');
INSERT 0 10000
postgres=*# rollback;
ROLLBACK
postgres=#
[dba@example5]$ cd ..
[dba@examplebase]$ cd ..
[dba@exampledb_data]$ vi postgresql.conf
[dba@exampledb_data]$ cd base
[dba@examplebase]$ cd 5
[dba@example5]$ ls -lrth 32789*
-rw-------. 1 dba dba  24K Mar 14 07:31 32789_fsm
-rw-------. 1 dba dba 440K Mar 14 07:34 32789
[dba@example5]$psql -p 5432 -U dba postgres
psql (17.4)
Type "help" for help.
postgres=# select * from pg_class where relname='arc';
postgres=# \q
[dba@example5]$ ls -lrth 32789*
-rw-------. 1 dba dba   0 Mar 14 07:58 32789_vm
-rw-------. 1 dba dba 16K Mar 14 07:58 32789_fsm
-rw-------. 1 dba dba   0 Mar 14 07:58 32789
[dba@example5]$psql -p 5432 -U dba postgres
psql (17.4)
Type "help" for help.
postgres=# select * from pg_stat_user_tables where relname='arc';
postgres=# \x
Expanded display is on.
postgres=# select * from pg_stat_user_tables where relname='arc';
postgres=# vacuum arc;
VACUUM
postgres=# select * from pg_stat_user_tables where relname='arc';
postgres=# \q
```

Basic structure of a table on disk

If you have a table called students, inside the database’s folder (like base/16384/), you’ll see something like this:

```
16384/12345       ← main data file (actual rows)
16384/12345_fsm   ← free space map
16384/12345_vm    ← visibility map
```

🧩 1. fsm — Free Space Map

Purpose: Tracks how much **free space** is available in each page (8KB block) of the table.

📖 **Why it exists:**

When PostgreSQL inserts a new row, it needs to know *where there's room*.
Instead of scanning the whole table file, it looks at the **fsm** to quickly find pages that have enough space.

Example: Let's say your table's data file (12345) has 4 pages:

Page 1 → 2000 bytes free
Page 2 → 1000 bytes free
Page 3 → 0 bytes free
Page 4 → 4000 bytes free

The FSM will store a small summary:

[Page 1: 25% free]
[Page 2: 12% free]
[Page 3: 0% free]
[Page 4: 50% free]

So when PostgreSQL needs to insert a row of 1500 bytes,
it checks FSM → finds Page 4 → inserts row there directly.
No need to scan every page!

📌 **Key point:**

- FSM = a *map of empty spaces* in data pages.
- Used by **INSERT** and **UPDATE** to reuse space efficiently.
- Helps reduce table bloat.

2. vm — Visibility Map

Purpose: Tracks which pages contain only **visible (fully committed)** tuples.
If all tuples on a page are visible to everyone, PostgreSQL marks it in the VM.

📖 **Why it exists:**

The **visibility map** helps PostgreSQL skip pages during VACUUM and ANALYZE.
If a page is fully visible (no dead rows), there's no need to scan it again.

Example: Imagine this:

Page 1 → All tuples committed → visible
Page 2 → Has some deleted rows → not visible
Page 3 → All tuples committed → visible

The VM will mark:

[Page 1: visible]
[Page 2: not visible]
[Page 3: visible]

When VACUUM runs, it will **skip Page 1 and 3**, saving tons of time.
It only scans pages that might need cleanup.

📌 **Key point:**

- VM = *map of which pages are safe to skip during vacuum*.
- Used by **VACUUM** and **index-only scans.**
- Greatly improves performance for large tables.

⚡ **Quick summary**

File	Name	Main purpose	Used by
tablename_fsm	Free Space Map	Find where to insert new rows (tracks empty space)	INSERT / UPDATE
tablename_vm	Visibility Map	Skip pages that don't need vacuuming or scanning	VACUUM / ANALYZE / Index-only scans

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